# CS5701 Quantitative Data Analysis (QDA) Suggested Pre-reads

Isabel Sassoon

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Here is a list of items to read and be familiar before the first lecture:

- It may be helpful to set up R and R Studio on your PC or laptop. You will have access to R and R studio in the Labs.
- If it helps check out these install instructions.
- The material in this document.

This document focuses on descriptive statistics for *numerical* data. Descriptive statistics for *categorical* data (such as frequency tables) are covered in the lecture and lab session this week.

## 1 Measures of Location

We will be using measures of location to describe data in the lecture and labs this week so below are some definitions and an opportunity to practice. We use the same small data set throughout this document - you'll get to check these very numbers against R's output when we reach the lab this week.

There are three commonly used measures of location:

- **Mode** the most frequent observation in the data set, there can be more than one
- **Mean** (average) this is calculated as the sum of the observations in the variable divided by the number of the observations.

  $$\bar{x} = \frac{1}{n}\Sigma_{i=1}^{n} x_i$$

  assuming we have an attribute $X$ which has the following observations $x_1, x_2, \ldots, x_n$
- **Median** when the set of observations is ordered from smallest to largest, the **median** is the value in the middle, or if there are an even number of observations then the mean of the two middle observations is used.

Note: the **mean** is affected by extreme values, the **median** is not.

### Exercise

Assuming this is our data set:

$$4, 7, 2, 5, 6, 2$$

Compute the following:

- **Mode**
- **Mean**
- **Median**

## Exercise - Worked Solution

Computing the measures of location this data set:

$$4, 7, 2, 5, 6, 2$$

- **Mode** The most frequent observation is: 2
- **Mean**
$$\bar{x} = \frac{1}{n}\Sigma_{i=1}^{n} x_i = \frac{1}{6} \times (4+7+2+5+6+2) = 4.333$$
- **Median** The values arranged from smallest to largest are: $2, 2, 4, 5, 6, 7$, there is an even number of values so the median is the mean of the two middle values $4, 5$. This results in 4.5

You'll compute these same values in R during the lab this week - check that `mean()` and `median()` give you the same answers (and if you attempt the stretch challenge, that your own mode function agrees too).

## Percentiles and Quantiles

- There are other measures of location, such as the *k-th percentile*. This measure finds the value that divides the data such that $k\%$ of the values are below it. The *75th percentile* and *25th percentile* are commonly used[1] - together they define the Inter Quartile Range, which we come back to below.
- The *50th percentile* is the median.

To compute the $k$-th percentile of $n$ sorted values:

1. Sort the $n$ data points in ascending order and give each one a rank from 1 to $n$.
2. Compute the position $p = 1 + \frac{k}{100} \times (n-1)$.
3. If $p$ is a whole number, the percentile is simply the value at that rank.
4. If $p$ falls between two ranks, interpolate between the two surrounding values in proportion to how far $p$ is between them.

This is the same method R's `quantile()` function uses by default, so a careful hand calculation and R's output should match.

[1] See p.67 of Crawley for more on percentiles.

**Worked example**

Using our data set $4,7,2,5,6,2$ (so $n = 6$), sorted this becomes $2,2,4,5,6,7$ (ranks 1 to 6).

**75th percentile:**

$$p = 1 + \frac{75}{100} \times (6-1) = 1 + 3.75 = 4.75$$

This falls between rank 4 (value 5) and rank 5 (value 6), three-quarters of the way from rank 4 to rank 5:

$$5 + 0.75 \times (6-5) = 5.75$$

**25th percentile:**

$$p = 1 + \frac{25}{100} \times (6-1) = 1 + 1.25 = 2.25$$

This falls between rank 2 (value 2) and rank 3 (value 4), one quarter of the way from rank 2 to rank 3:

$$2 + 0.25 \times (4-2) = 2.5$$

You'll compute these same numbers directly in R this week using `quantile(data_vector, c(0.25, 0.75))` - check that your answer matches.

**Weighted Mean - another measure of location**

There is another measure related to the mean which is useful when there are weights involved. A good example of this is when computing the overall grade point when different module grades have different weights as they account for a different number of credits (if you attended the induction talks this should sound familiar) .

$$\bar{x}_w = \frac{\Sigma_{i=1}^{n} w_i \times x_i}{\Sigma_{i=1}^{n} w_i}$$

Lets assume that you have 3 module grade points $13,16,12$ and that the respective weights (credits) are $15,15,30$ - what is the weighted grade point mean?

$$\bar{x}_w = \frac{13 \times 15 + 16 \times 15 + 12 \times 30}{15 + 15 + 30} = 13.25$$

# 2 Measures of Variation

There are different ways of quantifying the spread of a variable of data:

- The **range**: defined as *maximum value in the variable - minimum value*
- The **Inter Quartile Range - IQR** defined as the *75th Percentile - 25th Percentile*, this will measure the spread of the central 50% of the data
- The **Standard Deviation** - measures the spread of the observations from the mean

## Standard Deviation

The **standard deviation** is the average distance between each observation and the mean. There are two closely related versions, depending on whether you are working with an entire population or a sample.

When computing the standard deviation for a *sample* - which is almost always the case in practice, and what R's `sd()` function computes by default:

$$s = \sqrt{\frac{1}{n-1}\Sigma_{i=1}^{n}(x_i - \bar{x})^2}$$

The $n-1$ (rather than $n$) accounts for the fact that the mean itself was estimated from the same data before computing this.

For a complete *population*, the standard deviation is:

$$\sigma = \sqrt{\frac{1}{n}\Sigma_{i=1}^{n}(x_i - \bar{x})^2}$$

The square of the standard deviation is known as the **Variance** ($Var = s^2$ for a sample, or $\sigma^2$ for a population).

**Important:** R's `sd()` and `var()` functions compute the *sample* version by default. If you calculate the population version by hand, don't be surprised if it doesn't match R's output - check which formula you're using.

### Worked example

Using our data set $4, 7, 2, 5, 6, 2$, the mean is $\bar{x} = 4.333$ (see above). The squared deviations from the mean are:

| $x_i$ | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ |
|---|---|---|
| 4 | -0.333 | 0.111 |
| 7 | 2.667 | 7.111 |
| 2 | -2.333 | 5.444 |
| 5 | 0.667 | 0.444 |
| 6 | 1.667 | 2.778 |
| 2 | -2.333 | 5.444 |

Summing the last column: $\Sigma_{i=1}^{n}(x_i - \bar{x})^2 = 21.333$.

Sample variance: $s^2 = 21.333/5 = 4.267$, so sample standard deviation $s = \sqrt{4.267} = 2.066$.

Population variance: $\sigma^2 = 21.333/6 = 3.556$, so population standard deviation $\sigma = \sqrt{3.556} = 1.886$.

You'll compute these same numbers in R this week using `sd(data_vector)` and `var(data_vector)` - remember, R gives you the *sample* version (2.066 and 4.267), not the population one.

In Lecture and lab 1 we will use these concepts and see how they fit in Exploratory Data Analysis.